



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

AIRFLOW PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A MLOps/LLMOps

TOTAL: 10 QUESTIONS

Q1 What is Airflow and what AI/ML use case is it designed for? Model

Q6 How do you evaluate a model's performance in Airflow? Observability

Q2 What are the core abstractions in Airflow? Architecture

Q7 How do you deploy a model trained with Airflow? Lifecycle

Q3 How does Airflow handle training or inference? Configuration

Q8 How does Airflow handle distributed or multi-GPU training? Performance

Q4 How does Airflow manage data preprocessing? Hardening

Q9 How do you track experiments and versions in Airflow? Testing

Q5 How does Airflow support GPU/TPU acceleration? SSO / IdP

Q10 What are common pitfalls when using Airflow in production? Monitoring



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Airflow is a framework or service targeting

Mitigation

Airflow is a framework or service targeting a specific AI/ML domain training neural networks, building LLM pipelines, deploying models, or orchestrating agents. Understanding its scope helps you know when to use it vs alternatives.

A6 Evaluation uses a held-out validation set and

Observability

Evaluation uses a held-out validation set and metrics (accuracy, F1, BLEU, perplexity) appropriate to the task. Monitor both training and validation metrics to detect overfitting. Use a test set only at the very end to report final results.

A2 Airflow organizes work around abstractions like models, tensors, pipelines, agents, or embeddings. Learning these core types is the first step to using the framework effectively.

Primitives

A7 Export the model to a portable format

Automation

Export the model to a portable format (ONNX, TorchScript, SavedModel). Serve it via a model server (TorchServe, TF Serving, Triton) or embed it in an API endpoint. Monitor inference latency and drift in production.

A3 For training, Airflow defines a model architecture,

Hardening

For training, Airflow defines a model architecture, a loss function, and an optimizer. For inference, a trained model processes input data and returns predictions. Batch processing and hardware acceleration are key performance levers.

A8 Airflow provides data-parallel or model-parallel strategies to

Performance

Airflow provides data-parallel or model-parallel strategies to split work across GPUs. Data parallelism replicates the model on each GPU and averages gradients. Model parallelism splits layers across devices for models too large for one GPU.

A4 Airflow provides data loaders, tokenizers, or pipeline

Gotchas

Airflow provides data loaders, tokenizers, or pipeline components that transform raw data into the tensor or embedding format the model expects. Proper preprocessing is critical for model correctness and training speed.

A9 Use an experiment tracker (MLflow, Weights &

Assessment

Use an experiment tracker (MLflow, Weights & Biases, Comet) alongside Airflow to log hyperparameters, metrics, and artifacts. Track the code version and data version for full reproducibility.

A5 Airflow detects available accelerators and moves computation

Optimization

Airflow detects available accelerators and moves computation to them. Specify the device explicitly (cuda, mps, tpu) to avoid accidentally running expensive operations on CPU. Mixed-precision (fp16/bf16) further reduces memory and increases throughput.

A10 Common issues include training-serving skew (different preprocessing), missing input validation, no latency SLA enforcement, models not versioned, and no process to retrain on drifted data distributions.

Optimization

Common issues include training-serving skew (different preprocessing), missing input validation, no latency SLA enforcement, models not versioned, and no process to retrain on drifted data distributions.